



DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

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Year / Branch / Section: 2nd & Btech IT- 'A'

Semester: IV

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RAJALAKSHMI ENGINEERING
COLLEGE
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BONAFIDE CERTIFICATE

Name: V. CHANDRAN SUNDAR

Academic Year: ²⁰25-26 Semester: IV Branch: IT

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Certified that this is the bonafide record of work done by the above student in
the CS 23632 - Software Construction Laboratory

during the academic year 202 - 202

Signature of Faculty in-charge

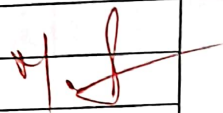
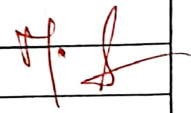
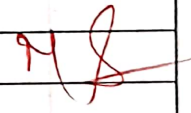
Submitted for the Practical Examination held on 12/5/26

Internal Examiner

External Examiner

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NAME: Chetan Sunda STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: 241001036

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Exp No: 1

Azure DevOps Environment Setup

Date: 21/1/26

Aim :-

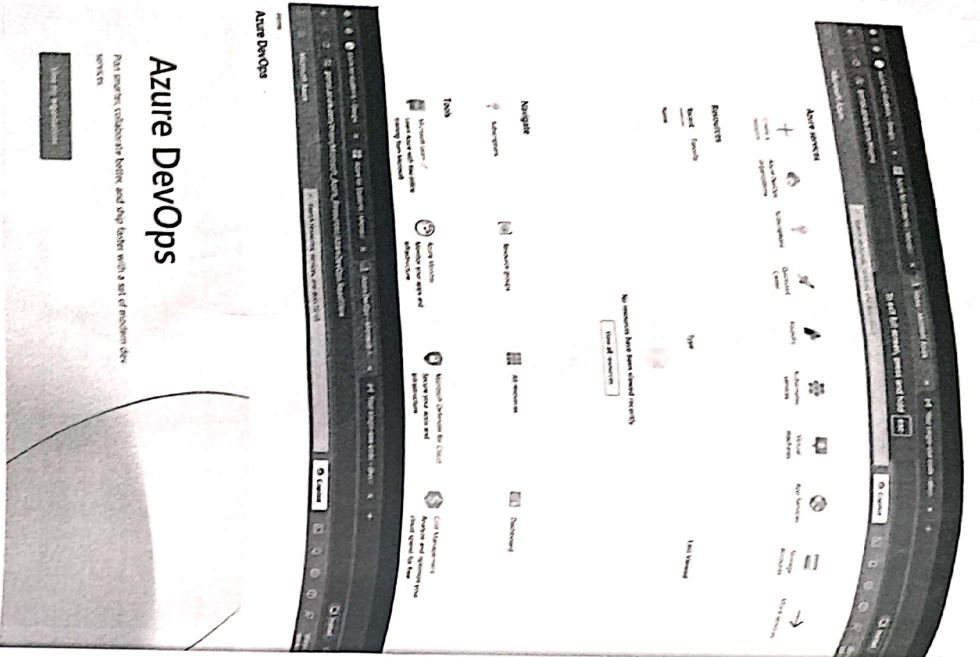
To set up and access the azure devops environment by creating an organisation through the azure portal.

Installation :-

1. Open your web browser & go to the Azure Website : <https://azure.microsoft.com/en-us/get-started/azure>.
Sign in using your Microsoft account credentials.
2. Azure home page
3. Open devops environment in the azure platform by typing azure devops in search box.
4. Click on the My azure devops organisation link & create an organisation & you should be taken to azure devops organisation home page.

Result:-

Successfully accessed the azure devops environment & created a new organisation through the azure portal.



Azure DevOps

Put together code, build, and deploy with a set of modern dev services

Start my organization

SERVICES

Explore the complete list of Azure DevOps services and learn how to use them. Includes links to documentation, pricing, and support.

Billing Management

Manage and Billing DevOps with your team. Access detailed billing information to understand your costs and optimize your usage.

- No: 2 Azure DevOps Project Setup and User Stories management.
- 28/1/20

Set up on Azure DevOps Project for collaboration agile work management.

Create an Azure account.

Create first project in your Organisation.

After the organisation is set up, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items & more.

In organisation home page, click on the New Project.

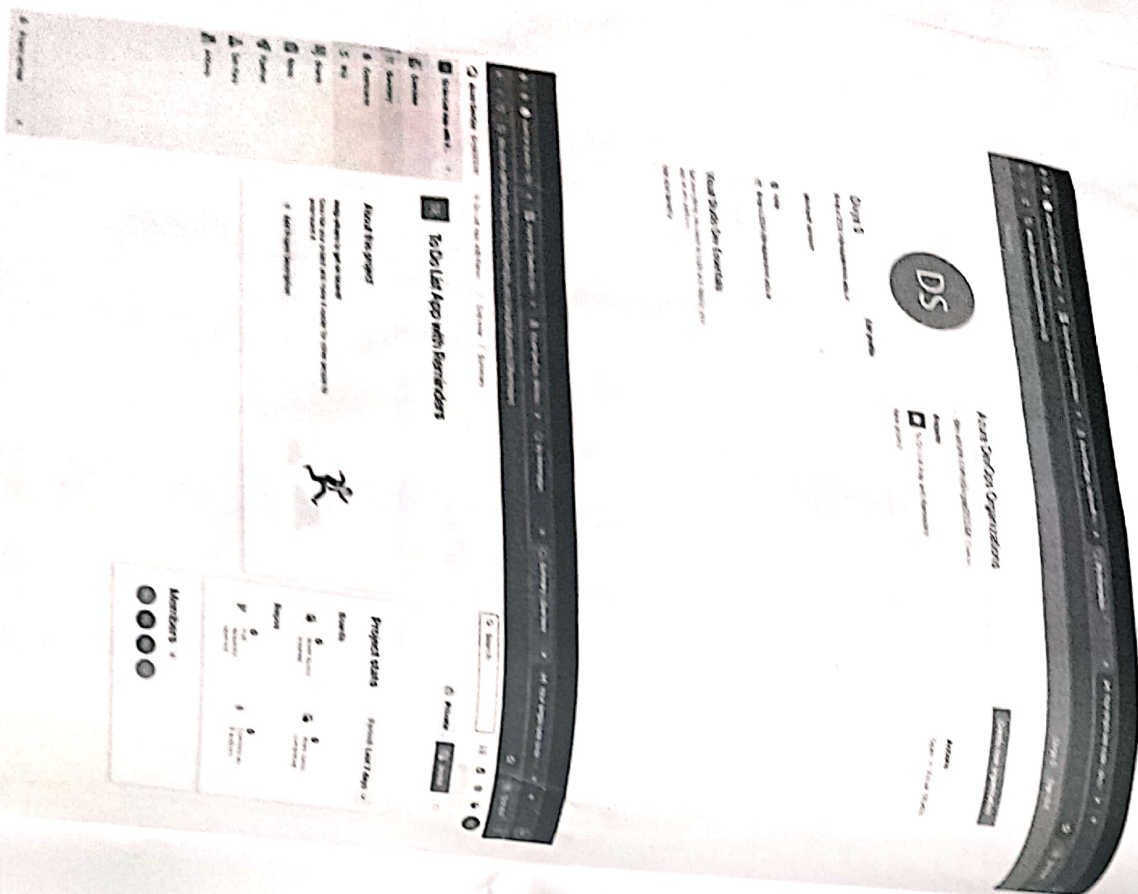
Enter your project name, description & visibility options.

Name :- To do list app to setup reminders.

Description :- Optionally, add a description to provide more input about Project.

Once you have filled out the details, click create at up your first project.

We logged in, ensure you are in the correct organisation. We're part of multiple organisation you can switch b/w them.



1. To manage user stories:

a) From left-hand navigation menu, click on Board. This will take you to the main Boards page, where you can manage work item, backlogs & sprints.

b) On the work item page, you'll see the option to Add a work item at the top. Alternatively, you can find a button or add new work item depending on the view you were in. From the add is work item dropdown, select user story.

Result:

~~Successfully created~~ an Azure devops project with user story management & agile workflow setup.

Exp NO.3 Setting up Epics, features &

Date:- 11/2/26 User stories

Aim :-

To learn how to create epics, features, user stories & tasks for a to-do list.

Procedure:-

~~Create Epics, features, user stories, Tasks.~~

User stories (main idea):-

As a user ~~set~~, I want to manage my daily tasks & set reminders.

1. Fill in Epics :-

My Epics :- Task Management & reminder system.

2. Fill in features:-

- Add new tasks
- Edit / update tasks
- delete tasks
- ~~set reminders~~
- Due date & time selection
- User friendly interface.

3. Fill in user story details :-

- View list of all tasks.
- Add a new task with title
- Set reminder for task
- Delete a task
- Receive notification

4. Detailed user stories

- As a user, I want to view all my tasks so that I can track what needs to be done.
- As a user, I want to add a new task with details so that I can organize my work.
- As a user, I want to set reminders for tasks so that I don't forget important activities.
- As a user, I want to edit or delete tasks so that I can manage changes.
- As a user, I want to receive notification so that I am alerted on time.

Result:-

Thus the ~~creation~~ of epic features, user stories & tasks for the to-do list / reminder application has been completed successfully.

Sprint Planning

Exp NO:- 4

Date : 25/2/26

Aim :-

To assign user stories to specific sprints for the To Do list / remainder app project planning.

Sprint : 1

~~Taskboard~~

8. View all Tasks

Assigned to : Charan

Iteration 1

9. Add new task (title & description)

Assigned to : Divya S

Sprint 2 :-

Taskboard

Iteration 2

10. Set remainder for tasks

Assigned to : Dinesh

11. Mark task as completed

Assigned to : Deepak

Sprint : 3

Taskboard

Iteration : 3

12. Edit / delete task

Assigned to :-

13. Receive ~~notification~~ alerts

~~Assigned to :~~

14. Display user - friendly interface

Assigned to :-

Result :-

The ~~sprint~~ are created for the To-Do list / remainder App project successfully.

11/3/26

scale poker estimation for user stories of the
 Db list / remainder app project.

or estimation:-

orig point measure :-

Effort

Complexity

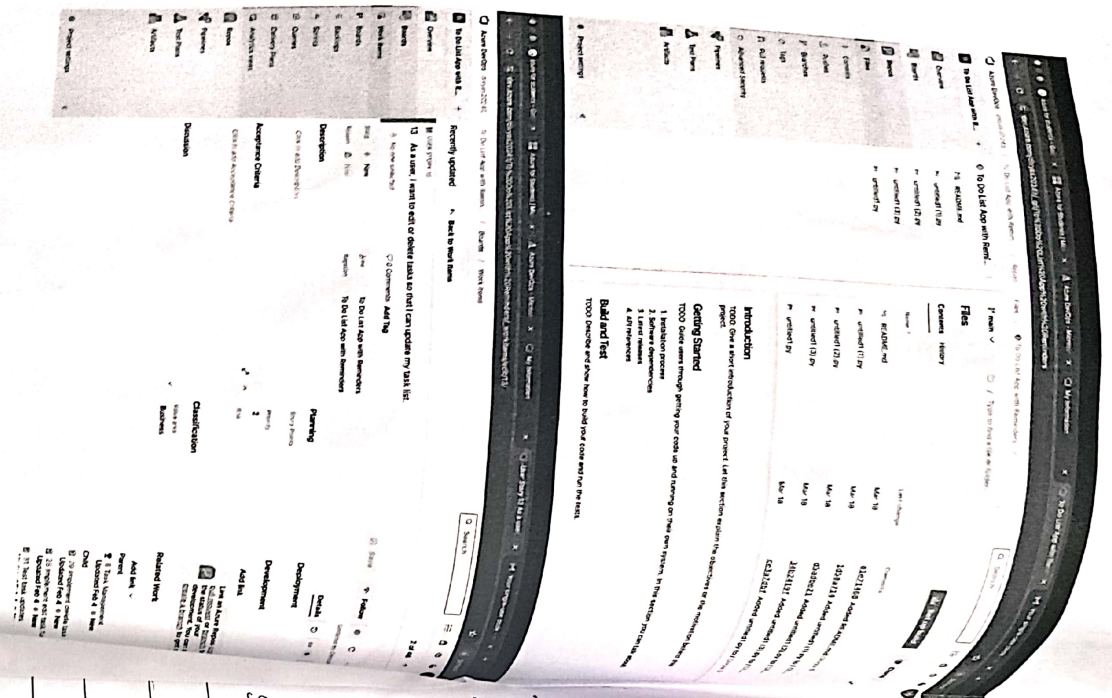
Time

we use Fibonacci series

1, 2, 3, 5, 8, 13, ...

story points for To-Do list / remainder app :-

User story.	story Point	Reason
Add new task	3	Input handling
View all task	2	Simple data display
Set remainder for task	5	Time scheduling
Mark task as completed	2	Simple status update
Edit / delete task	3	CRUD operation
Receive notification alert	5	Notification system
Display user-friendly interface	3	UI design & styling.



Total = 23 story points.

17

steps :-

Boards → Work item → User story → story points → save

4

Result:-

The estimation / story points are created for the
To -> list / remainder app project using poker
estimation successfully.

Exp No: 6

Designing class & sequence

Date :- 18/3/20

diagram for Project Architecture

Aim :- To design class diagram & sequence diagram for the To do list with reminder application

Objective :- The goal of this experiment is to understand how different components of application interact & how data flows between them using diagram

Classes :-

- User :- User Id, name, e-mail
- Task :- task Id, title, due Date, status.
- Remainder :- remainder Id, time.
- Notification :- message.

Relationships :-

- User → Task (1:M)
- Task → Remainder
- Remainder → Notification

Flow :-

1. User enters task
2. Task saved in system
3. User sets remainder
4. System schedule remainder
5. Notification sent

Epic, User stories, features :-

Epic :-

Task Management app with reminders.

Features :-

- Add / Edit / Delete Tasks.
- Set reminders.
- Notification.

User Stories :-

- User can create task
- User can set reminder
- User gets notification.

Result :-

~~Class~~

& sequence diagram were successfully designed.

Exp No : 7

Designing architectural & ER

Date :- 25/6/26

Diagrams for Project structure.

Aim :- To design an architectural diagram & ER diagram for To do list with reminder app.

1. Architectural diagram :-

Explanation :- The diagram represents the structure of the To do list application. It includes ;

- Frontend where user can add, edit, delete, logs & set reminders.
- Backend server :- Handles user requests & procedure tasks data.
- Notification service :- sends reminder to users at the scheduled time.

2. ER diagram :-

Explanation :- Diagram represents the database design of the To do list app.

- User Entity :- stores user information
- Tasks Entity :- stores task details.
- Reminder entity :- stores reminder details.

Relationships :-

- A user can have multiples tasks.
- Each tasks can have one/more remainder

Epic:- Develop To-do list App with remainder

Feature:- Add, Update, delete task, set remainder

User story:- User can create a task
User can set remainder
User gets notified on time.

Tasks:-

- Design UI
- Setup backend
- Connect database
- Implement remainder system

Result:- Thus, the architectural & ER diagram for To-do list with remainders app were successfully designed.

Exp No: 8 Testing - Test plan & Testcases

Date :- 1/4/20

Aim :-

To create test plan & test cases for the to-do list application with remainder functionality & validity system behaviour.

Test plan & Test cases :-

Test case design procedure :-

- ① Identify main features of system
- ② Define user action
- ③ Create happy path test cases
- ④ Assign test cases IDs.

User stories :-

- User can add task
- Can update or delete task
- Can set remainder for task
- Can make task as completed.

Test Cases :-

Test Suite :- Task Management

TC01 :- Add Task (Happy Path)

Action :- Enter valid task details.

TC02 :- Add Task (Error Path).

Test Suite :- Remainder Setup.

TC03 :- Set remainder (Happy Path)

Action :- Set value date & time for remainder

Result :- Remainder set successfully

TC04 :- Set remainder (Error Path)

Action :- Set date / time.

Result :- Error msg displayed.

TC05 :- Receive remainder (Happy Path)

Action :- Wait for schedule remainder time.

Result :- Notification received.

TC06 :- Mask Test complete (Error Path)

Action :- Try masking non-existing table

Result :- Error msg displayed.

Test Suite :- Delete Task

TC07 :- Delete Task (Error Path)

Action :- Delete already deleted table.

Result :- The test plan & test cases are created successfully for the To-do list application with remainder functionality, both happy path & error path.

Exp: 09

Load Testing & Pipelines

Date:

for Todo list to setup
Reminder

aim:- To perform load testing on a
Todo list app with reminder feature
& implement a pipeline for automation.

Load Testing - Description.

Simulate multiple users performing
actions like adding tasks & setting
reminders.

Steps:-

- 1) Login to Azure Portal
- 2) Create Azure load testing resource
- 3) Enter details (subscription, location)
- 4) Open resource → create test
- 5) Add http request

Observations:-

- Handles multiple users efficiently
- Response time stable.
- Reminder triggered correctly
- No major errors
- Good Scalability.

Pipeline Description

Automate build & testing of the Go-Do app using Azure DevOps.

Steps:-

- 1) Connect Github repo
- 2) Create Pipeline
- 3) Add YAML file

<1> YAML

trigger;

-main

pool;

vmImage: Ubuntu-latest

Steps:-

-task: Use python version

input:

version space: '3.x'

-script: pip install -r requirement.txt

-script: python -c "print('Pipeline Running')"

Observation:-

- Dependencies installed successfully
- Pipelines executed without error
- Logs verified execution
- Continuous integration

Result:-

Load Testing & pipeline Setup for todo list is completed successfully.



Exp 10, 10

Date:

Git Hub Project Structure & Naming Convention

Aim:-

To provide a clear & organized view of the project: folder structure & file naming convention, helping contributors & users easily understand, navigate & avoid the todo list app being reminder.

Git Hub Project Structure:-

1) Signup with Github

2) Go to setting

3) select repositories

4) click on create new repositories

5) & give name to repository

6) Make it public

7) Upload the folder from the system

8) then commit the changes

Git hub Project structure screenshots has been passed on left hand side.

Results :- The Github repository clearly displays the organized structure & consistent naming convention, making it easy for users & contributors to understand & navigate the codebase.